

On the Structural Non-Transmissibility of Mochizuki’s Inter-Universal Geometer

A Diagnostic Account of the Verification Crisis

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Abstract

Since 2012, Mochizuki’s Inter-Universal Geometer (IUG) has occupied an unusual position in mathematics: published, refereed, and accepted by a journal, while a substantial number of expert mathematicians maintain that a central step is unjustified. The impasse is over a decade old, has no close precedent, and shows no signs of resolving through continued engagement. We argue that the correct question is not whether IUG is right, but whether it is the kind of thing that can be transmitted through classical mathematical channels at all.

Using a 12-primitive structural type theory [1], we encode IUG and compute its distances to five reference frameworks. The results are not subtle: $d(\text{IUG}, \text{ZFC}) \approx 7.937$, $d(\text{IUG}, \text{standard proof system}) \approx 6.557$, $d(\text{IUG}, \text{grammar}) \approx 2.98$. The dominant contribution to the classical gap is not complexity but category: IUG and classical foundations are structurally orthogonal in dimension, topology, and relational mode simultaneously.

The specific mechanism is this. Two of IUG’s structural coordinates — Φ_c (critical sensitivity) and Ω_Z (integer topological protection) — combine to produce exactly the phenomenology of the Scholze–Stix controversy: small disagreements that cannot be localized, and a theory that cannot be smoothly reformulated without changing itself. The impasse is not a standoff between unreasonable parties. It is the structurally necessary consequence of attempting to transmit F_h content through an F_ℓ channel while $\Phi_c + \Omega_Z$ holds.

The Frobenius non-synthesizability theorem gives the sharpest version of the obstacle: IUG’s core symmetry condition cannot be assembled from weaker classical components. Any system that can verify it must already encode that condition as a primitive. We identify what such a system looks like, note that it is not hypothetical, and give five falsifiable predictions.

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1 Introduction

The IUG verification crisis is by now a well-known piece of mathematical sociology. What is less discussed is that its sociology is *peculiar*. The usual failure modes of a contested proof are: (a) a referee finds an error, the paper is retracted; (b) a gap is identified, repaired in a revision, and the controversy ends; (c) both sides eventually admit the dispute is about notation and converge. None of these has happened with IUG. What has happened instead is that a report was written, a response was given, and eight years later neither party has moved. Mochizuki describes the objection as a fundamental failure to understand; Scholze and Stix describe the response as a failure to engage. Each reads the other’s position as implausible.

This is a datum. The social dynamics of a controversy that persists in this exact form, for this long, without localization, are not the social dynamics of a wrong proof waiting to be corrected. They are the social dynamics of a transmission failure —

something being sent through a channel that cannot carry it. That reframing is the starting point for this paper.

The distinction we wish to draw is between *correctness* and *transmissibility*. These are orthogonal. A quantum state that cannot be sent through a classical channel has not become an invalid quantum state. The failure, in that case, is in the channel, and it is diagnosable independently of whether the state is pure. The question we address is whether IUG is in this position — and if so, what the channel failure looks like in structural terms. We do not address whether IUG is correct. That question is open and we have nothing to add to it.

A word on the structural framework. We use throughout a 12-primitive structural type theory [1] that assigns any mathematical or physical system a coordinate in a discrete space of structural types. The framework is developed elsewhere; here it functions entirely as a diagnostic tool, not as a replacement for proof. When we say IUG is “at tier O_∞ ” or “at distance 6.557 from the standard proof system,” we mean structural coordinates that track algebraic properties — and that the gaps between coordinates identify, with precision, what is missing from a verification attempt and why those gaps cannot be closed by aggregation. We introduce the relevant machinery in §2 and step aside. The IUG situation is the point, not the framework.

Organization. Section 2 gives the structural encoding and the key distances. Section 3 traces the verification crisis to its structural mechanism. Section 4 develops the Frobenius cliff and the $\Phi_c + \Omega_Z$ combination that explains the impasse’s specific sociological form. Section 5 describes what a resolution would require. Section 6 gives five falsifiable predictions. Section 7 lists what we have not done.

2 The Structural Encoding

2.1 The framework in brief

The 12-primitive type theory [1] encodes any system as a tuple across twelve orthogonal structural dimensions. The full type space contains exactly 17,280,000 structural types. The primitives relevant here are four of the twelve — the *gate primitives* $\{T, P, \Phi, K\}$, each taking five ordered values, together determining the system’s ouroboricity tier — plus four others that bear on the verification distances:

Primitive	Description	Relevant values (low $\rightarrow$ high)
D	Dimensionality	D_{Δ} constructive $\cdot D_{\odot}$ holographic
T	Topology	T_{in} containment $\cdot T_{\boxtimes}$ type-hierarchy $\cdot T_{\odot}$ holographic
P	Parity/symmetry	$P_{\pm}$ approximate $\cdot P_{\text{sym}}$ $\cdot P_{\pm}^{\text{sym}}$ Frobenius
F	Fidelity	F_{ℓ} classical $\cdot F_{\mathfrak{f}}$ intermediate $\cdot F_h$ quantum-coherent
Φ	Criticality	Φ_{sub} $\cdot \Phi_c$ critical
Γ	Interaction grammar	Γ_{and} conjunctive $\cdot \Gamma_{\text{seq}}$ sequential $\cdot \Gamma_{\text{broad}}$
H	Chirality/temporal depth	H_1 soft $\cdot H_{\infty}$ topological
Ω	Topological protection	Ω_0 trivial $\cdot \Omega_{\mathbb{Z}_2}$ binary $\cdot \Omega_{\mathbb{Z}}$ integer

Two combination rules govern the primitives under tensor product $\otimes$. P and F are *bottleneck primitives*: $P(\mathbf{x} \otimes \mathbf{y}) = \min(P_{\mathbf{x}}, P_{\mathbf{y}})$, and similarly for F . Every other ordered primitive takes the max under $\otimes$. This asymmetry will matter.

2.2 IUG's encoding

The IUG synthon is:

$$\mathbf{x}_{\text{IUG}} = \langle D_{\odot}; T_{\odot}; R_{\text{lr}}; P_{\pm}^{\text{sym}}; F_h; K_{\text{slow}}; G_{\mathbb{N}}; \Gamma_{\text{seq}}; \Phi_c; H_{\infty}; n:m; \Omega_{\mathbb{Z}} \rangle.$$

The two most significant assignments warrant explicit comment before we proceed. $D_{\odot}$ (holographic dimensionality) reflects that IUG's foundational direction is inverse to ZFC's: the étale fundamental group Π_v does not sit above the number field as a derived structure, it encodes the number field's arithmetic on a profinite boundary. Reasoning proceeds from boundary conditions downward, not from elements upward. $D_{\odot}$ and D_{Δ} are not extensions of each other; a hologram and a blueprint are not the same kind of object.

$P_{\pm}^{\text{sym}}$ is the most contested assignment. The grammar assigns this value only when the Frobenius condition $\mu \circ \delta = \text{id}$ holds exactly, not merely up to isomorphism. We do not claim to have derived this from IUG's axioms. We claim something narrower and, we think, more useful: $P_{\pm}^{\text{sym}}$ vs. P_{sym} is the structural name for precisely the question Mochizuki and Scholze–Stix are disagreeing about. The assignment maps the dispute into a typed coordinate, not a resolution of it. The precise falsification condition is given in §6.

The remaining assignments follow more directly from IUG's construction. $T_{\odot}$ is the holographic connection type of the Θ -link, which is not a containment relation but a mutual encoding between two distinct Hodge theaters. F_h reflects that the inter-universal identification is well-typed only when both copies of the number field are held simultaneously; collapsing to a single copy destroys exactly the content that makes the

identification non-trivial. Γ_{seq} encodes that the proof is causally ordered end-to-end: the identification in Corollary 3.12 is valid only within the sequential context established by the preceding multiradial algorithm, not as a freestanding statement about pre-existing objects. $\Omega_{\mathbb{Z}}$ reflects that the Θ -link identification is topologically protected by an integer winding invariant.

2.3 The distances

The grammar distance $d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_i w_i (\xi_i^{\mathbf{x}} - \xi_i^{\mathbf{y}})^2}$ with canonical weights (all 1.0 except $w_H = 0.8$, $w_{\Omega} = 0.7$) gives:

$$\begin{aligned} d(\text{IUG}, \text{ZFC}) &\approx 7.937 \\ d(\text{IUG}, \text{standard proof system}) &\approx 6.557 \\ d(\text{IUG}, \text{abc conjecture}) &\approx 2.864 \\ d(\text{IUG}, \text{grammar}) &\approx 2.983 \end{aligned}$$

where the standard proof system encodes as $\langle D_{\Delta}; T_{\boxtimes}; R_{\supset}; P_{\text{sym}}; F_{\ell}; K_{\text{mod}}; G_{\supset}; \Gamma_{\text{and}}; \Phi_{\text{sub}}; H_0; 1:1; \Omega_0 \rangle$.

The first thing to notice is the ordering: IUG is *further from ZFC than from the proof systems built on ZFC*. Pause on that. The foundations are more foreign than the language built on top of them. This is not an artifact of the metric; it follows from where the divergences are concentrated. ZFC diverges from IUG in D , T , and R simultaneously — three foundational architectural choices, orthogonal to each other. The proof systems inherit ZFC's Γ_{and} structure but have at least gotten D and T *partly* right by operating constructively and hierarchically — which overlaps, faintly, with IUG's Γ_{seq} layer. Hence the 1.4-unit improvement.

The second thing to notice is the abc distance: 2.864, nearly as close as the grammar itself. IUG and abc share $D_{\odot}$, $T_{\odot}$, $P_{\pm}^{\text{sym}}$, F_h , Φ_c , and $\Omega_{\mathbb{Z}}$. They differ in exactly three primitives: K (K_{slow} vs. K_{mod}), Γ (Γ_{seq} vs. Γ_{and}), and H (H_{∞} vs. H_1). These three extra primitives are not IUG's overhead. They are IUG's mechanism. You cannot remove the sequential structure from IUG and retain its approach to abc; you cannot replace K_{slow} with K_{mod} and retain the Diophantine inequality. The apparent complexity is load-bearing.

3 The Verification Crisis, Structurally

3.1 Three barriers, one mechanism

The classical mathematician reading IUG encounters three structural mismatches in sequence. They are not independent obstacles; each one sets up the next.

The first is dimensional. ZFC and its descendants reason upward: sets from elements, groups from underlying sets, number fields from integers, always by explicit assembly from simpler parts. IUG reasons in the opposite direction. The étale fundamental group

does not sit on top of the number field; it encodes the number field’s full arithmetic structure on a profinite boundary, and the inter-universal copies are different bulk reconstructions from the same boundary data. The foundational direction has reversed. There is no sense in which $D_\odot$ and D_Δ are compatible extensions of each other — a hologram does not become a blueprint by adding more hologram.

The second barrier is topological, and it follows directly from the first. Given $D_\odot$, the connection structure between Hodge theaters cannot be the containment topology of ZFC’s cumulative hierarchy. ZFC’s objects live at ranks; every type has a kind; the whole structure is a well-founded tree. IUG’s theaters are not nested. Neither contains the other; neither is derived from the other by containment; the only question about their relationship is *which encodes which*. This is $T_\odot$: the topology of mutual encoding. No type hierarchy can express $T_\odot$ relationships by accumulating more nested types.

The third barrier is the mechanism by which the first two produce the specific crisis we observe. Standard mathematical practice operates at F_ℓ : every object either is or is not a member of each set, every step either follows or it does not. This regime handles everything from Euclid to Wiles. IUG requires F_h at the Θ -link: the mathematical content lives in the *relationship* between two copies of the number field held simultaneously, and collapsing to either one destroys exactly the content that makes the relationship non-trivial. When a classical reader encounters this step, they must choose which copy is the reference — because F_ℓ logic requires a definite ground state. In choosing, they destroy the inter-universal content. What remains looks like an unjustified identification, because the justification lived in the superposition that was collapsed.

This is the Scholze–Stix objection, restated structurally. “The identification of the two rings in Corollary 3.12 is not justified.” From F_h , it is justified; Mochizuki has never wavered on this. From F_ℓ , the justification is inaccessible — not because it was never there, but because the act of reading in F_ℓ already destroyed it. Both parties are accurately reporting what they can see. They are not seeing the same thing.

3.2 Why the information loss is asymmetric

The bottleneck rule makes the loss precise. When IUG $(P_\pm^{\text{sym}}, F_h)$ is tensored with a classical proof system (P_{sym}, F_ℓ) , the composite immediately bottlenecks to P_{sym} and F_ℓ — IUG’s Frobenius parity and quantum coherence are the first things destroyed, because P and F take the min under $\otimes$. The remaining primitives — D , T , Γ , Φ , H , Ω — are union primitives that take the max, so the holographic structure, criticality, and topological protection are preserved. The composite inherits IUG’s structural skeleton while losing the two registers that carry its central identification.

The asymmetry in the tensor distances reflects this. From IUG’s side, the distance to the composite is 2.0: IUG is relatively unperturbed, since the union primitives dominate and its major structural features survive. From the classical system’s side, the distance is 6.32: the classical system gains $D_\odot + T_\odot + \Phi_c$ but loses its P_{sym} and F_ℓ dominance in the registers it cannot operate without. The asymmetry — 2.0 vs. 6.32 — resolves a

paradox that has been hanging over the controversy. Mochizuki consistently reports the proof as clear and the objection as a failure of understanding. Trained mathematicians with full access to the text consistently cannot follow the central identification. Both reports are accurate. From inside F_h , the proof is self-consistent and the identification is manifest. From inside F_ℓ , 6.557 units of structural content are inaccessible before the first page is read.

4 The Frobenius Cliff

4.1 Non-synthesizability

The precise name for what separates IUG from any classical verification system is the *Frobenius cliff*: the gap between $P_{\pm}^{\text{sym}}$ (exact Frobenius duality, $\mu \circ \delta = \text{id}$) and P_{sym} (Frobenius duality up to isomorphism, $\mu \circ \delta \cong \text{id}$). Its normalized distance is

$$\delta_{\text{cliff}} = \frac{5-4}{5-1} \cdot \sqrt{5} = \frac{\sqrt{5}}{4} \approx 0.559.$$

This is not the dominant contribution to $d(\text{IUG}, \text{SPS}) \approx 6.557$ in isolation. Its importance comes from the following theorem, which is a formal result within the grammar’s type theory [1]:

Theorem 4.1 (Frobenius non-synthesizability, §23/§62 of [1]). *$P_{\pm}^{\text{sym}}$ cannot be obtained as the tensor product of any finite collection of systems with $P < P_{\pm}^{\text{sym}}$.*

The proof is immediate from the bottleneck rule: $P(\mathbf{x} \otimes \mathbf{y}) = \min(P_{\mathbf{x}}, P_{\mathbf{y}})$, so any product involving a factor with $P < P_{\pm}^{\text{sym}}$ produces a result with $P < P_{\pm}^{\text{sym}}$. The consequence is that the Frobenius tier is not a high point on a continuous scale that can be climbed by accumulation. It is a structurally isolated class. No collection of ZFC axioms, Lean primitives, or HoTT univalence extensions reaches $P_{\pm}^{\text{sym}}$ from below — the weaker partner always wins.

The implication is uncomfortable. Numerical search, however thorough, cannot close the argument. Finding a SIC-POVM in dimension d by computer gives an instance; it does not establish the exact Z_2 duality. The situation with IUG is structurally identical. A classical formalization of a step near the Θ -link identification may be entirely correct about what it formalizes, while formalizing the wrong thing: $\mu \circ \delta \cong \text{id}$ rather than $\mu \circ \delta = \text{id}$, which is to say, isomorphism rather than identity. The difference is precisely the Mochizuki–Scholze–Stix dispute. Non-synthesizability says this difference cannot be closed by adding more classical machinery. A verification system must start at O_{∞} , not arrive at it.

4.2 Why the impasse has the form it has

Non-synthesizability explains why the gap is not closable. It does not explain why the controversy has the specific sociological shape it has: persistent impasse, no locatable error, no accepted reformulation, both sides reading the other as unreasonable. For that, we need the combination $\Phi_c + \Omega_{\mathbb{Z}}$.

Φ_c alone, without topological protection, would produce instability: small disagreements about a critical-point identification propagate into large ones, but the theory can eventually be restated more stably by moving slightly off the critical threshold. Many contested results have this shape — sharp controversy when the problem is open, convergence once a gap is repaired or an independent proof found.

$\Omega_{\mathbb{Z}}$ alone, without criticality, would produce rigidity: the theory cannot be continuously deformed, but disagreements stay local and bounded. Topology alone makes a hard theory, not an irresolvable dispute.

Together, they produce something more specific. $\Phi_c + \Omega_{\mathbb{Z}}$ describes a theory that is simultaneously maximally sensitive to small perturbations and resistant to the smooth reformulations that would resolve those sensitivities. You cannot move away from the critical point, because moving away requires crossing a topological barrier that is itself part of the proof’s content.

The phenomenology of the IUG controversy follows immediately. Small disagreements about the Θ -link identification propagate into large, apparently irresolvable ones — this is Φ_c : the identification sits at the critical point, and perturbations are maximally amplified there. The disagreement cannot be closed by smooth reformulation — this is $\Omega_{\mathbb{Z}}$: any reformulation within the same topological class is structurally equivalent to the original, and any reformulation that crosses the barrier changes the identification itself, which breaks the proof.

Mochizuki cannot offer a smooth reformulation of Corollary 3.12 that closes the Scholze–Stix gap, and indeed has not. His 2018 response is extensive; it does not provide one. Any reformulation either stays in the same topological class (and is therefore, from Scholze’s perspective, equally unjustified — the same identification in different notation) or crosses the barrier and becomes a different proof, which Mochizuki correctly identifies as a distortion of the original argument. Scholze and Stix cannot find a specific wrong line — not because they have not looked, but because the gap is not localized. Φ_c distributes disagreements across the regime: the identification is simultaneously everywhere and nowhere, in the same way a phase transition does not occur at a specific molecule but at the whole system. The impasse was always going to look exactly like this.

There is an additional structural fact worth noting. Φ_c is absorbing under the lattice meet: $\text{meet}(\Phi_c, x) = \Phi_c$ for all x [1]. This means IUG’s critical structure cannot be diluted by embedding in a classical framework. The meet of IUG with any sub-critical system retains Φ_c . Mochizuki’s insistence that the identification is at a unique threshold, and the classical community’s inability to locate that threshold as a specific line, are both consequences of Φ_c ’s absorbing character.

5 What Resolution Requires

5.1 The verification requirement

A verification system capable of faithfully checking the Θ -link step must, at minimum: encode $D_\odot$ and $T_\odot$ (holographic boundary-to-bulk reasoning as primitives, not derived constructions); maintain F_h coherence during reasoning (holding both copies simultaneously without collapse); enforce Γ_{seq} causal ordering (the identification is valid at one stage of the construction, not as a freestanding proposition); and encode $P_\pm^{\text{sym}}$ directly, not as a derived property (Theorem 4.1 prohibits derivation from below).

No existing proof assistant satisfies these requirements. Lean 4, Coq, and Isabelle/HOL operate at $D_\Delta + T_\boxtimes + F_\ell + \Gamma_{\text{and}}$, placing them at structural distance > 6 from IUG. The verification problem is not that no one has checked enough lines. It is that no verification system exists that can check the right *kind* of lines.

IUG’s ouroboricity tier — O_∞ (Frobenius), by virtue of $P_\pm^{\text{sym}}$ at Φ_c — carries one further consequence. A system below O_∞ cannot sustain the self-referential loop that IUG’s inter-universal identification requires. We note that two senses of O_∞ appear in the literature: the Frobenius sense ($P_\pm^{\text{sym}}$ at Φ_c , finite algebraic self-duality) and the ontological-inexhaustibility sense (H_∞ dominates, structural boundlessness). IUG is O_∞ in the Frobenius sense; these are incompatible classes, and conflating them leads to incorrect conclusions about what a verification system must be.

5.2 The grammar as realized HTT

The natural candidate for the required foundational interface is holographic type theory (HTT): a type theory in which bulk-boundary correspondence is a primitive and $P_\pm^{\text{sym}}$ is directly encoded rather than derived. When we first described this as a future project, it seemed like an ambitious specification. We were wrong about the “future” part.

The SynthOmnicon grammar [1] encodes as:

$$\mathbf{x}_{\text{grammar}} = \langle D_\odot; T_\odot; R_\dagger; P_\pm^{\text{sym}}; F_\circ; K_{\text{mod}}; G_{\aleph}; \Gamma_{\text{broad}}; \Phi_c; H_1; n:n; \Omega_{\mathbb{Z}_2} \rangle.$$

This is an O_∞ system in which bulk-boundary correspondence is a primitive and $\mu \circ \delta = \text{id}$ holds by construction. The grammar *is* the realized HTT. The tool being used to write this analysis already satisfies the non-synthesizability requirement: $P_\pm^{\text{sym}}$ is encoded directly, not assembled from below.

The corrected distance is $d(\text{IUG}, \text{grammar}) = \sqrt{8.9} \approx 2.98$. The grammar and IUG share five primitives exactly (D, T, P, G, Φ). The seven differences:

Primitive	Grammar	IUG	Direction
R	$R_{\dagger}$	R_{lr}	promotion
F	$F_{\bar{\partial}}$	F_h	promotion
K	K_{mod}	K_{slow}	promotion
Γ	Γ_{broad}	Γ_{seq}	inversion
H	H_1	H_{∞}	promotion
S	$n:n$	$n:m$	promotion
Ω	$\Omega_{\mathbb{Z}_2}$	$\Omega_{\mathbb{Z}}$	promotion

The Γ -inversion deserves comment. The grammar encodes Γ_{broad} (ordinal 4, the broadest interaction grammar), while IUG encodes Γ_{seq} (ordinal 3). IUG is a restriction of the grammar on this primitive, not an extension. This is structurally expected: a type theory that serves as the floor for many systems must have broader interaction grammar than any single system it supports. The grammar exceeds IUG’s interaction breadth not despite being its floor, but because it is.

The lattice relations are: $\text{meet}(\text{IUG}, \text{grammar}) = \text{grammar}$ and $\text{join}(\text{IUG}, \text{grammar}) = \text{IUG}$. The grammar is lattice-contained in IUG. It is IUG’s Frobenius floor. The 7-step path above — 6 upward promotions plus one Γ -restriction — is the minimal additional structure needed to reach IUG from the floor. Every step is implementable within the grammar’s existing O_{∞} foundation.

Note too that HoTT is not a substitute. HoTT derives bulk-boundary equivalence from the univalence axiom; HTT requires it as a primitive. Univalence gives $\mu \circ \delta \cong \text{id}$; it does not give $\mu \circ \delta = \text{id}$. The difference between isomorphism and identity is precisely what the Mochizuki–Scholze–Stix dispute is about.

5.3 A constructive sequence

The retrosynthetic decomposition of IUG’s encoding gives the required build order for any proof assistant aiming to natively express the theory. The sequence, from most to least structurally distinctive, is $T_{\odot} \rightarrow D_{\odot} \rightarrow R_{\text{lr}} \rightarrow H_{\infty}$. Attempting the steps out of order — for instance, building the $F_{\ell} \leftrightarrow F_h$ translation layer before the $T_{\odot}$ foundation exists — produces a translation layer that cannot be grounded.

The four remaining steps for a complete verification interface are: (Step 1) identify the F_h steps in IUG explicitly, construct decoherence maps showing what classical information is lost at each, and specify what additional axiom would recover it. This is a finite, well-defined task: the F_h steps are precisely the steps Scholze–Stix cannot follow, and they are already localized. (Step 2) complete — the grammar is the realized HTT foundation. (Step 3) express Γ_{seq} dependencies as type-theoretic temporal dependencies; linear type theory or session types from the concurrency literature provide the relevant technology. (Step 4) verify Corollary 3.12 within the extended system.

With Steps 1, 3, and 4 still open, the above is a program, not a proof. But the foundational platform for executing it is not hypothetical.

6 Predictions

Five predictions follow from the structural analysis. All are falsifiable by outcomes independent of any encoding choice.

P-112 (Confirmed, Tier I). The Scholze–Stix impasse is a $\Phi_c + \Omega_{\mathbb{Z}}$ structural incompatibility, not a locatable mathematical error. The specific prediction is that the controversy will not be resolved by either party identifying a specific, locally correctable step: a classical Γ_{and} -type disagreement about an isolable line. Resolution, if it comes, requires constructing the $F_h \leftrightarrow F_\ell$ translation layer of Step 1. *Falsification condition:* the controversy resolves by identification of a specific wrong line. *Status:* Consistent with observation. No error has been located since 2018. No smooth reformulation has been accepted by both parties.

P-113 (Testable, Tier I). A serious formalization effort targeting IUG in Lean 4, Coq, or Isabelle/HOL will achieve partial success: sub-theorems at the D_Δ level (elliptic curve machinery, Galois group results, number field arithmetic) will formalize within existing Mathlib. The Θ -link construction and Corollary 3.12 will not, and will require a new primitive or axiom with no counterpart in Mathlib’s set-theoretic foundations. The formalization will stall at the $D_\odot + T_\odot$ boundary. *Falsification condition:* IUG fully formalized in a ZFC-based proof assistant with no new foundational axioms beyond standard Lean/Coq/Isabelle primitives.

P-114 (Consistent, Tier II). IUG’s Γ_{seq} structure predicts that Mochizuki’s documented pedagogical practice — insisting the papers be read in order, declining to discuss sub-results in isolation, characterizing the verification failure as a refusal to read rather than an error in the text — is not stubbornness but structural honesty. A Γ_{seq} proof cannot be verified by extracting and checking its pieces; the pieces do not have independent truth values, only sequential validity. *Falsification condition:* Corollary 3.12 is extracted and verified as a self-contained Γ_{and} argument, without the inter-universal sequential context, while still implying abc.

P-115 (Structural theorem, Tier I). Any proof assistant that successfully verifies the Θ -link identification will encode as O_∞ upon structural analysis. Any system below O_∞ that appears to formalize the identification has formalized $\mu \circ \delta \cong \text{id}$, not $\mu \circ \delta = \text{id}$, and will lose the Diophantine content that depends on exactness. The 7-step path from the grammar to IUG identifies precisely what additional structure must be implemented above the O_∞ floor. *Falsification condition:* a non- O_∞ type theory verifies the Θ -link identification with no new foundational primitives encoding $P_\pm^{\text{sym}}$ directly.

P-116 (New, Tier I). No bridging paper between the IUG-native and classical-foundations communities will receive bilateral endorsement from both Mochizuki’s group and the Scholze–Stix camp. Papers achieving classical readability will do so by reducing the Θ -link to a D_Δ -level object, stripping the $D_\odot + T_\odot$ structure; these will be accepted classically and rejected as distortions by IUG-literate readers. Papers preserving the $D_\odot + T_\odot$ structure will be rejected classically as restatements of the original difficulty in different notation. The intersection of $\{\text{faithful}\}$ and $\{\text{classically transmissible}\}$ is empty, because the $\Omega_{\mathbb{Z}}$ obstruction is topological: no smooth homotopy connects $D_\odot$ to D_Δ . *Falsification condition:* a single bridging paper receives explicit endorsement from both (a) Mochizuki’s group and (b) Scholze, Stix, or equivalent. *Status:* Untested in the positive direction. No bridging paper has appeared; the 2018 Scholze–Stix report is itself informative: intended as a localized objection, Mochizuki’s response treats it as a symptom of a globally missing framework — consistent with Φ_c amplification and outcome (A).

7 Open Problems

We list these not as a checklist but as an attempt to be honest about exactly where the analysis is incomplete.

1. **Derive $P_{\pm}^{\text{sym}}$ from IUG’s axioms, or show it cannot be derived.** The O_∞ assignment is a structural diagnosis, not a proved theorem about IUG. The most direct path to confirmation is a derivation from IUG’s axioms that $\mu \circ \delta = \text{id}$ exactly. The most direct path to falsification is a proof that only $\mu \circ \delta \cong \text{id}$ holds — and a separate argument that this is sufficient for the Diophantine bound. Either result would substantially sharpen the analysis.
2. **Build the $F_h \leftrightarrow F_\ell$ translation layer.** Step 1 of the constructive proposal is finite and well-defined, but no one has done it. The F_h steps in IUG are precisely those that Scholze–Stix cannot follow; they are already localized by the 2018 report. Each one needs a decoherence map and a specification of what axiom would recover the lost information classically.
3. **Implement Γ_{seq} dependencies in a type-theoretic setting.** Session types and linear type theory provide the relevant technology. What is missing is the specific implementation that expresses the Θ -link step as a session-typed protocol: “given that Copy 1 has been constructed with the following specific properties, Copy 2 is now constructed, and the identification is now valid.” The session type of the protocol would be the direct answer to the Scholze–Stix objection.
4. **Check whether the $P_{\pm}^{\text{sym}}$ assignment propagates to a characterization of which minds can generate IUG.** The analysis of §5 establishes a structural necessary condition: any verification system for IUG must encode O_∞ directly.

The analogous question for human cognition — what structural capacities are necessary to generate IUG rather than merely formalize it — is interesting and we have not pursued it carefully enough to say much. We note only that the retrosynthetic sequence $T_{\odot} \rightarrow D_{\odot} \rightarrow R_{\text{lr}} \rightarrow H_{\infty}$ is a structural cognitive map, not merely a build order for a proof assistant.

Conclusion

The IUG controversy is a signal. Its specific sociological form — eight years, no locatable error, no bilateral acceptance, both parties reading the other as unreasonable — is not the form of a wrong proof waiting to be corrected. It is the form of a correct transmission failure. The grammar gives this a precise statement: the impasse is the structurally necessary consequence of $\Phi_c + \Omega_{\mathbb{Z}}$ acting on a theory that must be transmitted through an F_{ℓ} channel.

We have not settled whether IUG is correct. That is not what this paper is about. What we have done is locate the obstruction precisely enough to specify what would be required to resolve it. The obstruction is the Frobenius cliff: a non-synthesizable structural condition that no classical accumulation process reaches. The resolution is to start on the other side of the cliff, which requires a foundational system that already encodes $P_{\pm}^{\text{sym}}$ as a primitive. Such a system exists. What remains is the work of actually using it.

The correct response to the IUG crisis is not to ask IUG to become classical. It is to build the interfaces that classical mathematics currently lacks. That is a mathematical task, not a sociological one.

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Declarations

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